UNIX FILE COMMANDS

In this module following important file handling commands will be discussed:

- touch
- cat
- cat > (single redirection)
- cat >>(double redirection)
- cp
- mv
- cmp
- comm
- diff

touch command

Touch command is used in Unix to create an empty file.

```
$ touch file1 file2
$ ls -l
-rw-r--r-- 1 user group 0 2014-02-08 7:14 file1
-rw-r--r-- 1 user group 0 2014-02-08 7:14 file2
$
```

Touch command is used to change the timestamps (access time and modification time of a file).

cat command

Use of cat command:
'cat' command is used to display a file content.
Syntax:

$ cat filename [filename]

```
$ cat data
This is an unix file
This is line two
$
```

cat > and cat >> command

Redirection operator (> and >>) can be used with cat command to take input from standard input device(keyboard) and store them in a file.
Syntax: $ cat > [filename]

```
$ cat > data
The file created using
cat > filename
^d
$
```

cat with single redirection(cat >filename) , creates a new file if no file with the given file name already exists and overrites the file if a file with given file name already exists.

cat with double redirection (cat >>filename) appends the content given through standard input to the file.

cp command

cp command is used to copy one file's content to another file.
Syntax: $ cp [source filename] [destination filename]
Switches:
1. cp -i [sourcefilename] [destination filename] This command copies the content of a file interactively.

2.cp -r [source_directory] [destination_directory] This command copies the whole file hierarchy.

3. cp -v [sourcefilename] [destination filename] Copies in verbose mode, explains what is being done

mv command

mv command is used for:
1. for renaming a file in same directory.
2. for moving a file to a different directory.
Syntax:
$ mv [sourcefilename] [destinationfilename]
$ mv [source_directory] [destination_directory]

Directory Commands:

Every item in the UNIX filesystem tree is either a file, or a directory. A directory is like a file folder. A directory can contain files, and other directories. A directory contained within another is called the child of the other. A directory in the filesystem tree may have many children, but it can only have one parent. A file can hold information, but cannot contain other files, or directories.

The place in the file system tree where an user is located is called the **current working directory** (which can be listed using **pwd** command).

As a user logs in, user is put in his/her **home directory.**

mkdir(make directory) command:

The mkdir command creates a directory with specified name in the present working directory or specified path.
Syntax: $ mkdir [ ]

ls Command

Lists the contents of any directory (current directory if no directory specified).
Sorts entries without option
$ ls [option(s)] [filename]

File comparison commands:

cmp – This command is used to compare two files, and if they differ, tells the first byte and line number where they differ. If input files are same , returns nothing.

Syntax :

cmp options file1 file2

To use cmp commands both the files need to be sorted.

comm- This command is used to compare two sorted files.

Syntax:

comm [options] file1 file2

- One set of options allows selection of 'columns' to suppress.
  - -1: suppress lines unique to file1 (column 1)
  - -2: suppress lines unique to file2 (column 2)
  - -3: suppress lines common to file1 and file2 (column3)

Example: To show only lines common between file1 and file2, command is

$ comm -12 file1 file2

diff - This command is used to compare two files line by line. The output indicates how the lines in each file are different, and the steps that will be involved in changing file1 to file2.

The change commands are in the format [range][acd][range]. The range on the left may be a line number or a comma-separated range of line numbers referring to file1, and the range on the right similarly refers to file2. The character in the middle indicates the action i.e. add, change or delete.

- 'LaR' – Add lines in range 'R' from file2 after line 'L' in file1.
- 'FcT' – Change lines in range 'F' of file1 to lines in range 'T' of file2.
- 'RdL' – Delete lines in range 'R' from file1 that would have appeared at line 'L' in file2

Syntax: diff [options] file1 file2